

Tailorable Force-Sensing Skins for Manipulators Using Textile and Additive Manufacturing

Adam Imdieke¹, Heidi Woelfle², Brad Holschuh² and Karthik Desingh¹

Abstract—Robotic manipulation in human-centered and cluttered environments requires not only precise end-effector control but also the ability to sense and regulate contact forces across the entire arm. Restricting tactile sensing to the fingertips or gripper limits a robot’s ability to detect, modulate, and safely respond to contact occurring along its links during motion. Whole-arm tactile sensing enables both contact avoidance and deliberate contact exploitation, yet practical tactile skins remain difficult to realize due to trade-offs between fabrication cost, surface coverage, sensing density, usable force range, and long-term repeatability. We present two complementary whole-arm tactile skin architectures that jointly address this trade-off: a textile-based design configured for lower-force sensitivity and contact avoidance, and a 3D-printed conformal shell configured for higher-threshold contact embracing. Both designs use a resistive taxel matrix and cost under \$10 per link. We provide fabrication guidelines, tunable design parameters, and quantitative characterization of force range, repeatability, and hysteresis. Experiments on a mobile manipulator demonstrate both contact avoidance and controlled contact-embracing behaviors in real-world tasks. Additional quantitative characterization data and supplementary results are available at the project website: <https://rpm-lab-umn.github.io/TactileSkin/>

I. INTRODUCTION

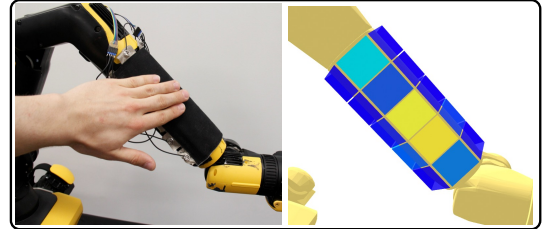
Humans naturally exploit whole-arm contact during manipulation, using surfaces to brace, stabilize, or guide movements in cluttered environments. Recent advances in robotic tactile sensing have demonstrated the value of extending sensing beyond grippers to enable safer physical human-robot interaction [1], whole-body contact awareness [2], [3], contact-rich manipulation [4], [5], and force-feedback teleoperation [6]. Despite these advances, widespread adoption remains limited: most robotic systems still lack tactile coverage beyond the hands, limiting their ability to detect unexpected arm contact, maintain contact-force targets, or use environmental contacts during manipulation [7], [8].

Several challenges continue to hinder the broader adoption of whole-arm or whole-body tactile sensing in robotics. Existing research prototypes that extend tactile sensing to the forearm or whole arm [7], [8], [11] often rely on custom PCBs, proprietary sensor modules, or specialized fabrication processes, creating a gap between demonstration systems and reproducible deployment.

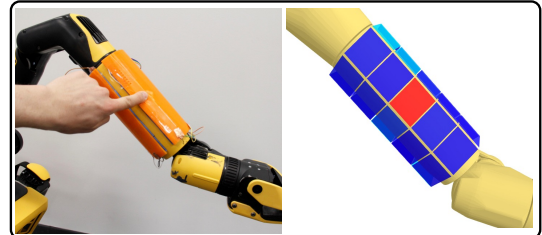
Table I compares the sensing layouts, reported measurements, and manufacturing requirements of representative systems. The reported sensing layout and measurement preserve

¹Robotics: Perception and Manipulation (RPM) Lab, Department of Computer Science & Engineering, University of Minnesota, Minneapolis, MN, USA.

²Wearable Technology Lab (WTL), Department of Design Innovation, University of Minnesota, St. Paul, MN, USA.



(a) Textile-based skin sensor on the robot arm and corresponding force readings.



(b) Additive-manufacturing-based skin sensor on the robot arm and corresponding force readings.

Fig. 1. Textile and AM sensors for a mobile manipulator robot (Boston Dynamics Spot), enabling force-sensitive manipulation across different contact-force requirements.

each work’s native reporting convention while distinguishing calibrated force or pressure from relative magnitude signals, contact localization, and binary contact.

To bridge these gaps, a tactile skin must satisfy both sensing and deployment requirements: it must detect the contact forces relevant to manipulation, produce repeatable readings under repeated loading, and be manufacturable for existing robot links without permanent modification. We therefore pursued the research question of *which measurable properties make tactile skins useful for mobile manipulators, and how can those skins be fabricated using textile and additive manufacturing (AM) methods?* Through iterative design, we evaluate three factors for whole-arm or body sensing: (1) **operating force range** - the range of forces a sensor can distinctly sense, (2) **sensing repeatability** - the consistency of sensor measurements across repeated loading and unloading cycles, (3) **deployability** - fabrication and installation requirements on commercially available robots without permanent link modification.

We propose two complementary sensor architectures for manufacturing force-sensing skins with taxels arranged on a 20 mm grid, material cost under \$10 per link, and fabrication paths based on ironing fabric layers or printing flexible

TABLE I

COMPARISON OF WHOLE-ARM TACTILE SENSING SYSTEMS USING QUANTITIES AND CAPABILITIES REPORTED BY EACH SOURCE.

<i>Whole-Arm Tactile Sensor</i>	<i>Sensing Performance</i>	<i>Manufacturing Requirements</i>
Related Works	Reported Sensing Layout and Measurement	Required Manufacturing Equipment
Maiolino et al. [7]	10 pressure taxels per module; calibrated pressure.	Custom FPCB, thermoformed fabric
CushSense [1]	3 cm $\times$ 3 cm taxels; normal-force characterization over 0–40 N, with sensing reported up to 55 N.	Cutting/sewing, CDC boards
RobotSweater [9]	16–64 taxels per prototype; relative force signal with a ~ 2.5 N threshold.	Industrial knitting machine
Johnson et al. [10]	More than 8,000 taxels across a one-square-meter prototype; relative pressure signal at > 50 FPS.	Fabric assembly, custom SPI PCB
CySkin [8]	11 pressure sensors per module; pressure readings at up to 20 Hz.	CySkin modules and IHBs
GenTact [3]	30 nodules across six skins; binary contact localization.	Multi-material 3D printer
Kohlbrenner et al. [11]	64 capacitance intersections; contact localization with 5.7 ± 3.0 mm error.	3D printing, silicone molding, Muca board
Cheng et al. [2]	62–1,260 cells per deployment; force and proximity signals.	Custom cell PCBAs, silicone molding
Ours		
Textile sensor (ours)	20 mm taxel pitch; calibrated normal force.	Iron, Velostat, snap buttons
AM sensor (ours)	20 mm taxel pitch; calibrated normal force.	3D printer, Velostat, copper tape

shells. Our design instances of these two architectures differ in force sensing range, flexibility, and repeatability, so we configure them for two mobile manipulation behaviors: contact avoidance and contact embracing. The first is a textile-based sensor constructed from conductive fabric layers with a piezoresistive material sandwiched between them. The second is an additive-manufacturing-based sensor composed of 3D-printed components shaped to match the robot arm’s geometry, forming a pressure-sensing matrix of conductive layers separated by a piezoresistive layer and generated directly from the arm’s 3D mesh.

Figure 1 shows both sensor implementations mounted on the Spot arm.

The contributions of this paper are as follows:

- Two complementary sensor architectures for whole-arm force-sensing skins, instantiated as a textile-based and an additive-manufacturing-based design, each with taxels spaced 20 mm center-to-center, material cost under \$10 per link, and removable mounting on a commercially available mobile manipulator.
- Fabrication guidelines and sensor characterization that demonstrate the practical feasibility, force-sensing range, and repeatability of each design, along with the key design parameters that can be tuned to meet application-specific requirements.
- Experimental validation on a mobile manipulator platform demonstrating real-world sensing performance across contact avoidance and contact embracing scenarios.

II. RELATED WORK

Existing whole-arm and whole-body tactile sensing systems vary considerably in their sensing outputs and fabrication requirements. Early systems extended tactile coverage beyond the gripper using custom capacitive PCBs or proprietary modules [7], [8]. More recent systems use sewn capacitive taxels, custom multimodal cells, mutual-capacitance intersections, machine-knitted skins, or printed sensing nodules [1]–[3], [9], [11]. Table I compares these

systems using only quantities and capabilities explicitly reported by their authors.

Compton et al. [12] demonstrated a layered textile FSR by placing piezoresistive fabric between knit electrodes, and Golgouneh and Dunne [13] further developed a soft e-textile matrix for on-body force sensing. These works show that a row–column textile stack can provide distributed force measurements, but they target wearable sensing rather than integration on rigid robot links. Our textile sensor adapts this principle using cut Velostat, ironed conductive-fabric electrodes, snap-button connections, and a removable sleeve characterized on the target manipulation platform.

RobotSweater [9] demonstrates that textile tactile skins can be deployed on mobile manipulators. Its three reported prototypes contain 16, 48, and 64 taxels and provide force-correlated tactile readings, but fabrication requires an industrial knitting machine. Our fabric sensor addresses the same whole-arm deployment problem using a cut-and-ironed construction made from sheet materials, while exposing material choices that can tune the sensing response.

Kohlbrenner et al. [3] introduced GenTact, a conformal tactile skin fabricated using multi-material printing and conductive filaments. Its FR3 demonstration used 30 sensing nodules across six skin units, and its RC-delay sensing reports binary contact rather than calibrated force magnitude. Our AM sensor separates structure from sensing: the printed shell provides conformal mounting and a designed normally-open gap, while copper tape and Velostat form a 20 mm-pitch resistive matrix for force estimation.

Maiolino et al. [7] developed a flexible large-area tactile skin using custom capacitive PCBs and supporting electronics. CySkin [8] similarly offers whole-arm tactile sensing through proprietary sensor modules, interface boards, and the Skinware stack. In contrast, our fabric sensor uses off-the-shelf fabric, Velostat, and snap-button wiring, with material cost under \$10 per link in our implementation.

Kohlbrenner et al. [11] used 64 mutual-capacitance intersections embedded in silicone on a 3D-printed base and trained a model to localize contact on a semi-conical surface.

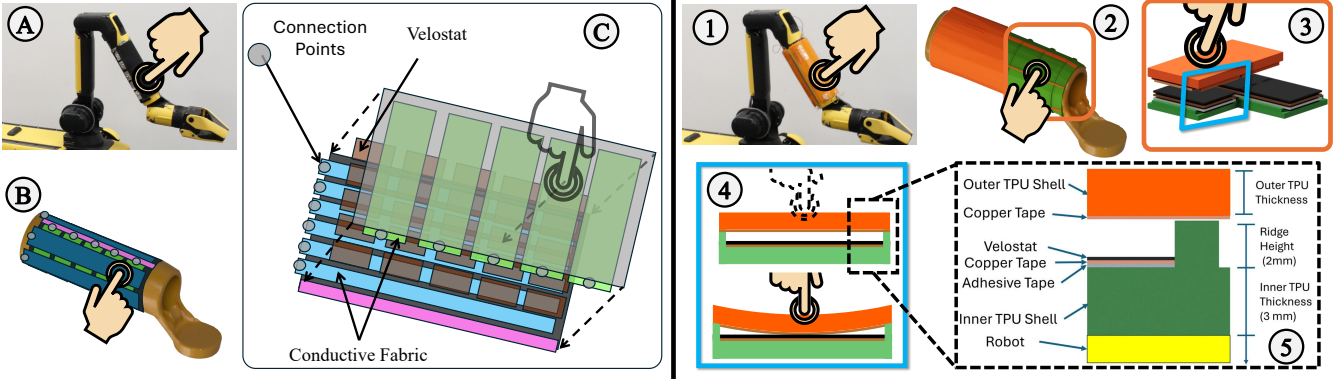


Fig. 2. Illustration of our tactile skin sensor designs (best viewed in color). **Left:** Textile Sensor A) showing the sensor on the Spot Robot arm, B) showing the sensor on the joint, C) showing the layers of the sensor. **Right:** AM Sensor 1) showing the sensor on the Spot Robot arm, 2) showing the inner and outer layers of the sensor along with their ridges, 3) showing the 3D structure of the taxel array, 4) showing the 2D cross-section before and during contact, 5) showing the layer structure of each taxel.

Our AM sensor instead uses a row-column resistive matrix, so contact location follows directly from the active taxel without a learned localization model. Cheng et al. [2] combined force and proximity sensing using custom taxel PCBAs and distributed electronics, while CushSense [1] used sewn $3\text{ cm} \times 3\text{ cm}$ capacitive taxels. Our AM design uses robot-link meshes to generate a conforming shell with a 20 mm taxel pitch and material cost under \$10 per link.

Together, these systems combine different sensing outputs, fabrication processes, electronics, and mounting strategies. Our two designs contribute accessible alternatives: a fabric sleeve configured for lower-force contact detection and an AM shell configured for conformal mounting and higher-threshold contact embracing.

III. SENSOR DESIGNS

Here we describe the sensor designs for the two architectures, tailored to the arm links of the Spot mobile manipulation platform. We design them to balance three practical requirements: (1) operating force range, (2) sensing repeatability across repeated load cycles, and (3) deployability on commercial robots through removable link-mounted skins rather than permanent link modification.

Both sensor designs share a common operating principle: a matrix of conductive rows and columns separated by a piezoresistive layer (Velostat). When contact occurs at a taxel, the piezoresistive layer compresses and becomes more conductive, changing the resistance at the row-column intersection, which is measured by a microcontroller through a voltage divider circuit. Both designs target link-level coverage, analog force readings across individual taxels, and fabrication with commercially available materials; the AM workflow uses robot-link meshes to adapt the shell geometry to a target platform.

We implement two complementary variants: (1) a textile-based sensor using conductive fabric layers, and (2) an additive manufacturing (AM) based sensor with 3D-printed conformal shells and copper-tape electrodes. We chose design

variables such that the textile sensor prioritizes flexibility and lower-force contact response, while the AM sensor provides a conformal shell for higher-threshold contact. In these configurations, they support the two behaviors evaluated in Sec. VI: contact avoidance and contact embracing. We describe the design and fabrication of each in the following sections, with an overview shown in Fig. 2.

A. Textile Sensor Design

The textile sensor shown in Fig. 2 is a flexible sleeve that mounts to the robot arm without rigid shells. Both the textile and AM sensors use a matrix of conductive rows and columns separated by a piezoresistive Velostat layer, with the same taxel pitch of approximately 20 mm. The sensor is formed into a cylindrical sleeve using hook-and-loop tape as a closure, allowing it to be wrapped around and removed from the robot link for maintenance. Rows and columns are interfaced with the microcontroller via snap buttons.

- 1) **Inner layer:** The inner layer template is derived from a 2D projection of the robot link geometry and cut from a textured off-the-shelf fabric, chosen for flexibility and to reduce slipping against the robot surface. Rows of conductive fabric are adhered to the inner layer, with excess fabric removed at taxel boundaries to electrically isolate adjacent rows.
- 2) **Outer layer:** The outer layer uses the same template and also uses off-the-shelf fabric, providing a soft contact surface for improved sensitivity. Columns of conductive fabric are adhered to the outer layer, with excess fabric removed at taxel boundaries to isolate adjacent columns.
- 3) **Force Sensitive Layer:** Velostat squares are cut such that they are slightly wider than the conductive rows and columns to prevent short-circuiting.
- 4) **Assembly:** The inner layer, Velostat squares, and outer layer are stacked in order and a fabric adhesive is used around each Velostat square to secure it. The sensor is then wrapped around the robot link, then secured with hook-and-loop tape. Snap buttons connect wires to each

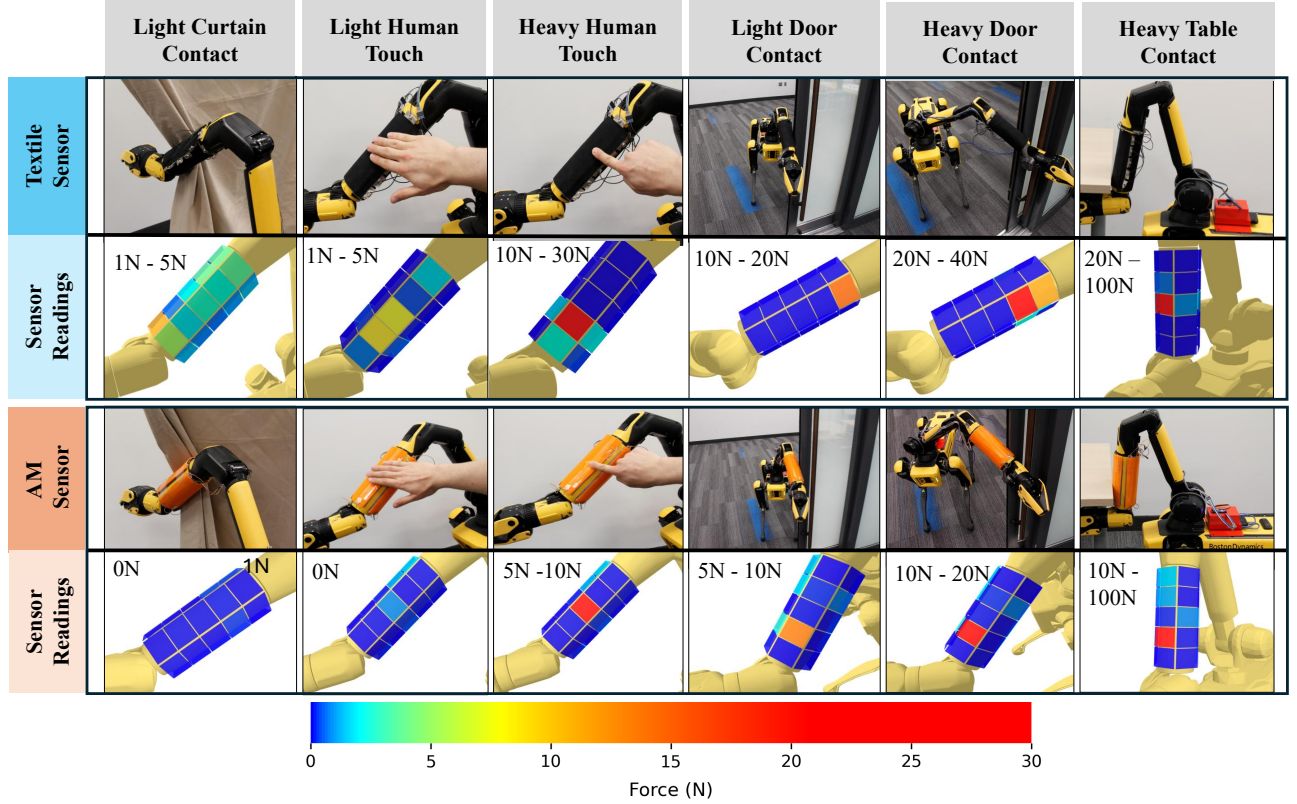


Fig. 3. Real-world deployment of the textile and AM sensors across representative manipulation scenarios. For each scenario, the middle panel visualizes tactile responses during contact, and the annotated values report the corresponding force ranges estimated from the sensor models. Upper bounds above 30 N are extrapolations beyond the 30 N calibration maximum and are included only to indicate the observed high-force regime.

row and column, which are routed to a microcontroller. The fabrication of this sensor requires fabric cutting, ironing, adhesive, and snap-button wiring, and the material cost is less than \$10 per Spot arm link, excluding shared readout electronics.

B. Additive Manufacturing (AM) Sensor Design

The AM sensor is intended for tasks where direct textile wrapping is less appropriate: the skin must hold a repeatable shape on curved rigid links, avoid pre-compressing the sensing layer during installation, and respond primarily after a deliberate contact threshold is reached. A simpler taped or fabric-wrapped matrix can provide contact sensing, but its baseline response depends on wrap tension and local curvature. The printed shell instead fixes the electrode spacing and contact geometry relative to the robot link.

To design the 3D-printed skin, we start from the link-level mesh files provided by Boston Dynamics (Fig. 2-2, yellow) as part of their URDF models. The AM sensor uses the same 20 mm center-to-center taxel spacing as the textile sensor, but places the Velostat/copper-tape matrix inside a 95A TPU shell so that contact must first close a designed gap. The selected geometry was chosen to satisfy three constraints rather than to optimize over the full design space: maintain the same taxel spacing as the textile prototype, provide enough shell thickness and clearance for printable

95A TPU link covers, and create a mechanical threshold suitable for the contact-embracing task in Sec. VI. This geometry targets higher-threshold contacts than the textile sleeve while preserving link-level conformability. For each link, we apply the following sequence of CAD operations:

- 1) **Inner layer:** Generate a conformal shell with a uniform thickness of 3 mm offset outward from the robot’s base geometry (Fig. 2-2, green).
- 2) **Outer layer:** A second shell (3 mm thick) offset 5 mm from the robot’s base geometry to accommodate the sensing stack (Fig. 2-2, orange).
- 3) **Ridges:** Within the inter-shell region, add 1 mm-thick horizontal ridges on the inner shell and vertical ridges on the outer shell, each 2 mm high. The spacing between adjacent ridges determines the taxel density (Fig. 2-2, cutout).
- 4) **Clearance cuts:** Material is removed near kinematic joints to prevent interference during motion, and a longitudinal cut is added along each shell to facilitate installation on the robot arm (Fig. 2-1).
- 5) **Assembly:** Print shells in 95A TPU. Laminate Velostat with copper tape (conductive adhesive), adhere to the inner shell, and apply copper tape to the outer shell for opposing electrodes. Connect rows/columns to the microcontroller and secure the assembled shells to the robot arm.

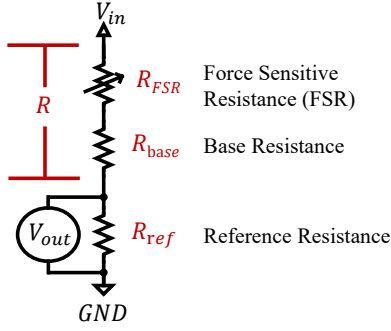


Fig. 4. We model both of our sensors as a piezoresistive element in series with a base resistance. To measure the resistance of each element, we use a voltage divider and measure the voltage across a reference resistor. V_{in} and GND are GPIO from a microcontroller, where both are set to high-impedance after sampling to reduce cross-talk.

The fabrication of this sensor requires a 3D printer capable of printing flexible filaments, copper tape, Velostat, and wiring, and the material cost remains under \$10 per Spot arm link, excluding shared readout electronics. In the characterization protocol of Sec. V, both sensors are tested over repeated 0–30 N load–unload cycles to measure operating force range, repeatability, and hysteresis.

C. Design Variables

Both architectures expose practical design variables that can be adjusted to match task requirements for force range, repeatability, and fabrication effort. If only binary contact is required, the Velostat layer can be omitted; continuous force estimation requires retaining and calibrating the piezoresistive layer.

For the textile sensor, the principal variables are outer-layer stiffness and thickness, optional spacer material, and sleeve tension. Based on the sensor structure, softer or thinner outer layers are expected to shift the response toward lower contact forces, while thicker or stiffer fabrics should shift it upward; these effects were not independently swept in this study. A spacer material such as foam may broaden the response range, while minimizing the spacer may reduce hysteresis. Installation tension can also change pre-compression and the sensor’s characteristic curve across deployments. The characterized configuration is configured for lower-threshold contact avoidance.

For the AM sensor, shell thickness, ridge height and spacing, material stiffness, and infill density determine the printed contact geometry. We characterize one configuration rather than a full parameter sweep. Lower shell stiffness, lower infill, or shorter ridges are expected to reduce the force needed to close the designed gap, while the opposite changes should increase the contact threshold. Ridge spacing sets taxel pitch, which is 20 mm in our prototype. TPU viscoelasticity and print structure may affect hysteresis, while the rigid-conformal shell reduces sensitivity to installation tension. Unlike the textile sensor’s normally contacting stack,

the AM sensor’s designed gap closes only under compression, configuring it for higher-threshold contact-embracing tasks.

IV. SENSOR MODELING

We model each sensor as a grid of taxels, where each taxel provides an analog reading corresponding to the magnitude of contact force at that location. The taxels are arranged in a grid pattern, with rows and columns defined by the conductive materials on the inner and outer layers of the sensor. We calibrate the sensor by applying known normal forces at taxel centers and recording the corresponding sensor readings, allowing us to map taxel resistance to contact force. The readout circuit is shown in Fig. 4.

To measure the resistance of each taxel, we use a voltage divider circuit, where the taxel is connected in series with a known reference resistor. The voltage across the reference resistor is measured, and we first use Ohm’s law to calculate the resistance R of the taxel based on the measured voltage V_{out} , the input voltage V_{in} , and the known reference resistor value R_{ref}

$$R = \left(\frac{V_{in} - V_{out}}{V_{out}} \right) R_{ref} \quad (1)$$

To account for the non-force-dependent resistance of conductive traces and contact interfaces in series with the piezoresistive material, we write the total resistance as

$$R = R_{base} + R_{FSR} \quad (2)$$

This model prevents the estimated force from diverging by representing the sensor’s asymptotic convergence to its hardware-defined resistance floor at high loads. We use a power law to model the FSR:

$$R(F) = AF^B + R_{base} \quad (3)$$

Solving for force gives:

$$F = \left(\frac{R - R_{base}}{A} \right)^{\frac{1}{B}} \quad (4)$$

where A , B , and R_{base} are power-law parameters determined through calibration and curve fitting. This model allows us to estimate contact force at each taxel from measured resistance.

V. SENSOR CHARACTERIZATION

We characterize both sensor architectures using an Instron testing machine. We use a 1000Ω reference resistor for the AM sensor and a 47Ω reference resistor for the textile sensor in a voltage-divider circuit. For each taxel, we apply a 7.5×7.5 mm square applicator at the taxel center and record the voltage response over four load–unload cycles from 0 to 30 N. We fit a power law with base resistance, $R(F) = AF^B + R_{base}$, to loading and unloading responses and use the fitted curves for performance analysis and force estimation. To evaluate performance, we report operating force range and consistency metrics. The project website provides supplementary taxel-level calibration curves, load–unload traces, and quantitative summaries.

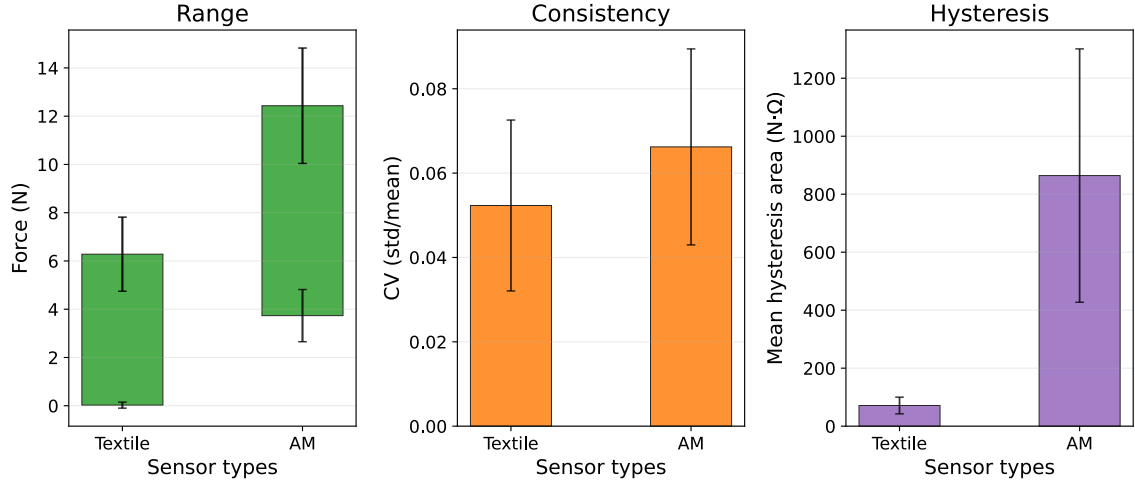


Fig. 5. **Left:** The *Operating Force Range* visualizes the force interval between the minimum detectable force (detection threshold) and the saturation force, which defines the effective range of forces that can be reliably measured by the sensor. **Middle:** The *Consistency* plot shows the multi-force repeatability metric, which captures the variability of sensor readings across repeated loading cycles at different force levels. **Right:** The *Hysteresis* plot illustrates the difference between loading and unloading curves, indicating the degree to which the sensor’s response depends on its loading history. Together, these plots provide insights into the performance characteristics of the textile and AM sensors.

A. Operating Force Range

We define the operating force range of each sensor as the force interval from its detection threshold (minimum detectable force) to saturation (maximum force for which changes in resistance remain informative). Fig. 5 summarizes the resulting coverage across sensor variants.

1) *Minimum Detectable Force:* In our pipeline, the detection threshold is measured per taxel as the *break force*:

$$F_{\text{det}} = \min\{F > 0 : R(F) \leq R_{\text{thr}}\}, \quad R_{\text{thr}} = 4000 \, \Omega,$$

2) *Saturation Force:* The saturation force is measured from the fitted response $R(F) = AF^B + R_{\text{base}}$ by mapping it through the expected voltage-divider readout

$$V(F) = V_{\text{in}} \frac{R_{\text{ref}}}{R(F) + R_{\text{ref}}}, \quad V_{\text{in}} = 5 \, \text{V},$$

and defining

$$F_{\text{sat}} = \max\left\{F \in [F_{\text{det}}, 30 \, \text{N}] : |dV/dF| \geq 0.03 \, \text{V/N}\right\},$$

where R_{ref} is the reference resistor value, which is selected over a range of reference resistors to find the value that maximizes the sensitive range. We find the optimal value of R_{ref} is approximately R_{base} . The sensitive range is $[F_{\text{det}}, F_{\text{sat}}]$: below F_{det} the sensor behaves like an open circuit (negligible signal), while above F_{sat} the response flattens and force estimates become ambiguous.

3) *Findings:* Across taxels, the textile sensor operated over approximately 0–6 N, whereas the AM sensor operated over approximately 4–12 N (Fig. 5, left). Their mean multi-force CV values were approximately 0.052 and 0.066, respectively, indicating comparable cycle-to-cycle repeatability. The greater AM hysteresis is consistent with deformation of the viscoelastic TPU shell. Together, these results support using the textile configuration for early contact detection and the AM configuration for higher-threshold interactions.

These intervals characterize the tested mechanical configurations and reference resistors rather than intrinsic material limits; changing the mechanical stack or readout resistor will shift the measurable range.

The AM sensor’s designed gap explains its elevated detection threshold: as force increases, shell deformation closes the gap and produces a more abrupt resistance transition.

When wrapped on the robot arm and tightened with hook-and-loop fastening, the effective sensitive range of Velostat-based sensors decreases relative to flat characterization. In our tests, mounting primarily reduced the textile sensor’s maximum force range. We attribute this reduction to pre-compression and non-uniform pressure distribution across curved surfaces; Velostat bending sensitivity may also shift baseline resistance and response behavior [13].

B. Consistency

1) *Repeatability:* For each taxel, we sample 10 points uniformly across $[F_{\text{det}}, 30 \, \text{N}]$, and compute the standard deviation of the sensor readings across repeated loading cycles at each force level. We summarize single-force repeatability as the standard deviation $\sigma_R(F)$ of the sensor readings across cycles at each force level F :

$$\sigma_R(F) = \text{std}\left(\{R_i(F)\}_{i=1}^{N_c}\right), \quad (5)$$

along with the corresponding mean $\mu_R(F)$ and the number of valid cycles N_c .

To normalize the repeatability metric across different resistances and sensor configurations, we compute the coefficient of variation (CV) across cycles at each force level F :

$$\text{CV}(F_j) = \frac{\sigma_R(F_j)}{\mu_R(F_j)} \quad (6)$$

We average the CV across each sampled force level to obtain a single multi-force repeatability metric for each taxel:

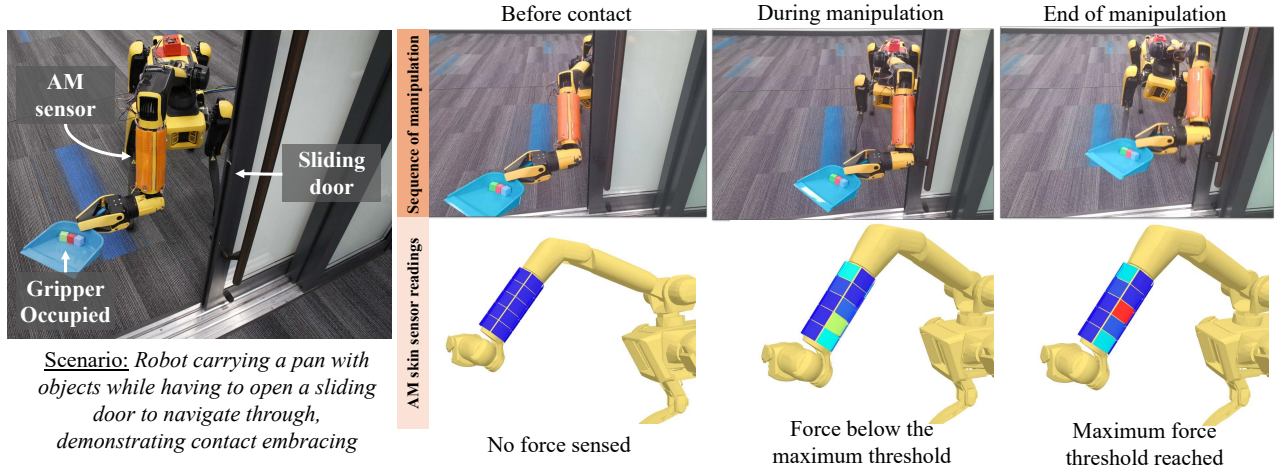


Fig. 6. The contact embracing task aims to balance a dustpan in the gripper, and open a door while preventing large forces. To open the door, we create a new end effector on the arm of the robot and move along its normal until a force above a threshold is achieved. We also optimize the rotation angle of the gripper to keep the dustpan level.

$$\overline{CV} = \frac{1}{N_s} \sum_{j=1}^{N_s} CV(F_j) \quad (7)$$

and we then report the mean $\overline{CV}$ across taxels, along with the corresponding standard deviation across taxels, to summarize the overall repeatability of the sensor. This multi-force repeatability metric captures the consistency of the sensor's response across its operating range and the variability across taxels, providing a comprehensive measure of the sensor's repeatability performance.

2) *Hysteresis*: We quantify hysteresis using the discrepancy between the loading and unloading curves in the resistance domain. For each taxel, we fit power-law models to the loading and unloading data,

$$R(F) = AF^B + R_{\text{base}}, \quad (8)$$

and invert them to obtain $F_{\text{load}}(R)$ and $F_{\text{unload}}(R)$ over the region where inversion is valid. Hysteresis is then computed as the area between the two inverted curves:

$$H_{\text{area}} = \int_{R_{\min}}^{R_{\max}} |F_{\text{load}}(R) - F_{\text{unload}}(R)| dR, \quad (9)$$

This area-based metric captures cumulative hysteretic behavior across the operating range, rather than at a single point.

3) *Findings*: The greater AM hysteresis (Fig. 5, right) is attributed primarily to viscoelastic effects in the printed shell; the textile sensor shows less separation between loading and unloading curves under these test conditions.

VI. REAL-WORLD USE CASES

Given the characterization results, we use the AM sensor for contact-embracing tasks that use its designed contact threshold and rigid-conformal shell, while we use the textile sensor for contact-avoidance tasks that use its lower measured detection threshold.

We use our sensors to perform two whole-arm contact behaviors on the Boston Dynamics Spot platform. Figure 3

summarizes the contact-force regimes observed across representative interactions. We developed a custom contact-aware whole-body inverse kinematics algorithm that incorporates tactile feedback from our sensors to enable both contact avoidance and contact embracing behaviors during manipulation.

A. Contact Embracing

We demonstrate the ability of the AM sensor to enable contact with rigid objects in the environment through a contact embracing task, where the robot uses tactile feedback to open a door while balancing a dustpan in the gripper (Fig. 6). We create a new end effector on the arm of the robot, and move along its normal until a force above a threshold is achieved. We also optimize the rotation angle of the gripper to prevent the dustpan from tipping over.

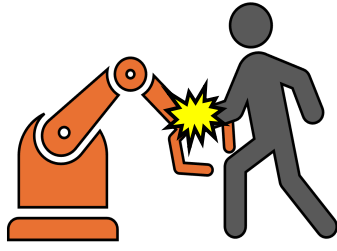
This task requires the robot to actively seek contact with the door, sense when the door is fully open, and maintain a desired contact force during the motion.

B. Contact Avoidance

The textile sensor has a lower detection threshold than the AM sensor in our characterization, so we use it for contact avoidance tasks where the robot needs to react to light contacts and avoid collisions. We demonstrate this in a contact avoidance task (Fig. 7), where the robot reacts to unexpected contacts while maintaining its primary end-effector goal. In this task, the robot uses tactile feedback from the textile sensor to detect contact and optimize the inverse kinematics solution to move away from the contact while still maintaining the end-effector XYZ position.

VII. CONCLUSION

In this work, we presented two complementary whole-arm tactile skin architectures designed to address trade-offs between fabrication complexity, cost, and sensing performance in mobile manipulation. By leveraging textile and additive manufacturing (AM) processes, we demonstrated



Scenario: *Robot encountering human contact during manipulation and initiating a contact avoidance maneuver.*

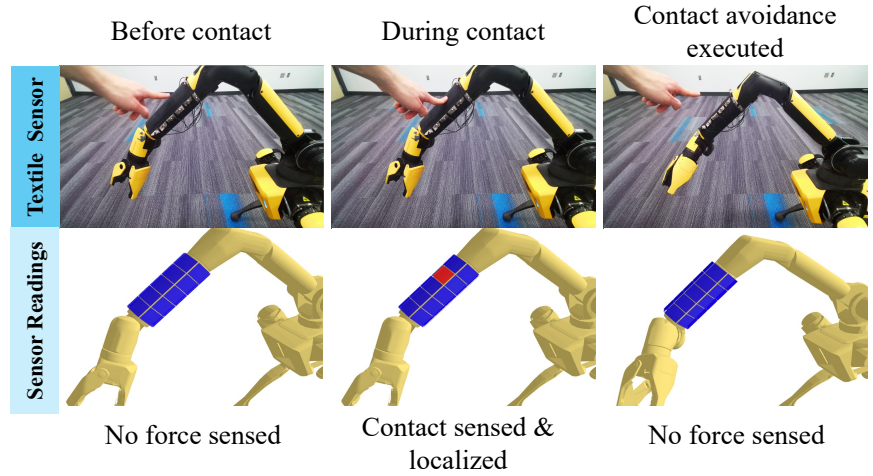


Fig. 7. Textile-based contact avoidance maneuver, where the textile skin detects human contact. The robot then utilizes a contact-aware inverse kinematics algorithm to move the arm link away from the contact while maintaining its XYZ position.

force-sensing skins with taxels spaced 20 mm (2 cm) center-to-center and material cost under \$10 per link using Velostat, conductive fabric, copper tape, and TPU.

Our characterization revealed distinct operational profiles for each design: the textile-based sensor, with its lower detection threshold and flexible sleeve, is configured for contact avoidance and light-touch human-robot interaction. In contrast, the AM-based sensor's rigid-conformal shell and "normally open" taxel resistance are configured for contact-embracing tasks, such as bracing against surfaces or modulating higher-force interactions. Real-world experiments on a mobile manipulator validated these designs in contact avoidance and contact embracing behaviors, showing that tactile feedback can be integrated into whole-body inverse kinematics for contact-aware motion.

Despite these gains, specific technical challenges remain for future investigation. The current taxel matrix scales roughly $2\sqrt{N_{\text{taxels}}}$ in wiring complexity, and rotary axes make external wiring difficult. Future work in embedded local processing and wireless communication could reduce wiring issues at scale. Additionally, while our power-law model captures force-resistance mappings, compensation for hysteresis—particularly in the viscoelastic TPU shells of the AM design—and temperature-induced drift will be important for tasks requiring tighter force-estimation error. Finally, future iterations will explore and characterize the design and material variables for both sensors. By providing open fabrication guidelines and quantitative characterization, this work establishes a baseline for whole-body tactile awareness in contact-rich mobile manipulation.

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